

Project 3 Phase 1: State Estimation using Extended Kalman Filter

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1 Methodology

State and Prediction. The filter estimates the 15-dimensional state $\mathbf{x} = [\mathbf{p}, \boldsymbol{\theta}, \mathbf{v}, \mathbf{b}_g, \mathbf{b}_a]^\top$ (position, Euler angles, velocity, gyro/accel biases). The continuous dynamics use bias-compensated IMU readings $\boldsymbol{\omega} = \boldsymbol{\omega}_m - \mathbf{b}_g$, $\mathbf{a} = \mathbf{a}_m - \mathbf{b}_a$:

$$\dot{\mathbf{p}} = \mathbf{v}, \quad \dot{\boldsymbol{\theta}} = \boldsymbol{\omega}, \quad \dot{\mathbf{v}} = \mathbf{R}(\boldsymbol{\theta})\mathbf{a} + \mathbf{g},$$

with $\mathbf{g} = (0, 0, -9.81)$ and bias random walks $\dot{\mathbf{b}}_g = \mathbf{n}_{bg}$, $\dot{\mathbf{b}}_a = \mathbf{n}_{ba}$.

Discrete prediction uses forward Euler for the mean, and a first-order covariance propagation:

$$\mathbf{x}_{k+1}^- = \mathbf{f}(\mathbf{x}_k, \boldsymbol{\omega}_k, \mathbf{a}_k, dt), \quad \mathbf{P}_{k+1}^- = \mathbf{F}_d \mathbf{P}_k \mathbf{F}_d^\top + \mathbf{U} \mathbf{Q} \mathbf{U}^\top dt,$$

with $\mathbf{F}_d = \mathbf{I} + \mathbf{F}dt$ and $\mathbf{Q}$ diagonal containing IMU noise densities.

Visual Measurement. The tag detector provides the camera pose in the world frame. Using the extrinsic calibration (camera-to-IMU translation $(0.05, 0.05, 0)$ m, rotation $\mathbf{R}_c^i = \text{diag}(1, -1, -1)$), we convert it to the IMU frame:

$$\mathbf{p}_{\text{IMU}}^w = \mathbf{p}_{\text{cam}}^w - \mathbf{R}_{\text{cam}}^w \mathbf{p}_{\text{cam}}^{\text{IMU}}, \quad \mathbf{R}_{\text{IMU}}^w = \mathbf{R}_{\text{cam}}^w (\mathbf{R}_c^i)^\top.$$

The measurement residual $\mathbf{y} = \mathbf{z} - \mathbf{H}\hat{\mathbf{x}}^-$ (position and Euler angles) is computed; yaw is wrapped to $[-\pi, \pi)$.

Mahalanobis Gating. To reject outliers, we check the squared Mahalanobis distance $D^2 = \mathbf{y}^\top \mathbf{S}^{-1} \mathbf{y}$ with $\mathbf{S} = \mathbf{H} \mathbf{P}^- \mathbf{H}^\top + \mathbf{R}_t$. If $D^2 > 16.81$ (99% $\chi^2(6)$), the measurement is discarded. Otherwise, the standard EKF update is applied: $\mathbf{K} = \mathbf{P}^- \mathbf{H}^\top \mathbf{S}^{-1}$, $\mathbf{x}^+ = \mathbf{x}^- + \mathbf{K} \mathbf{y}$, $\mathbf{P}^+ = (\mathbf{I} - \mathbf{K} \mathbf{H}) \mathbf{P}^-$.

2 Results

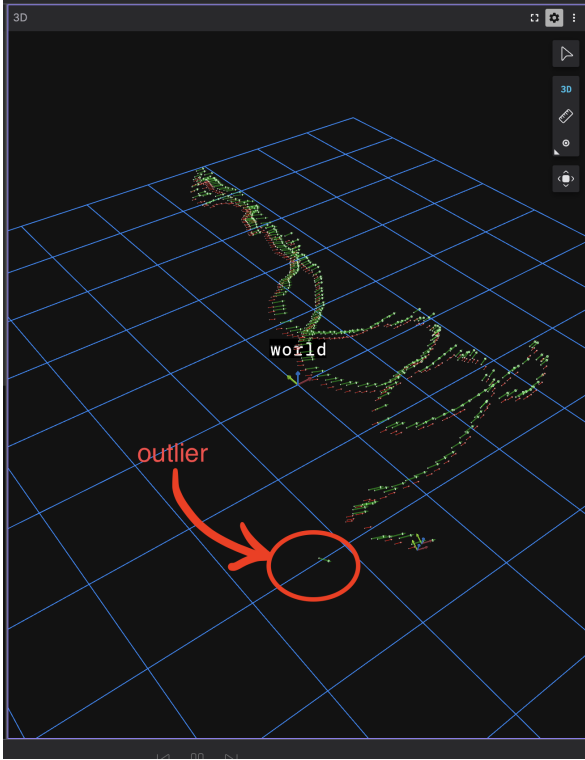


Figure 1: Trajectory in Foxglove: red arrow = EKF odom (`/ekf/ekf_odom`), green arrow = raw tag detection (`/tag_detector/odom_ref`).

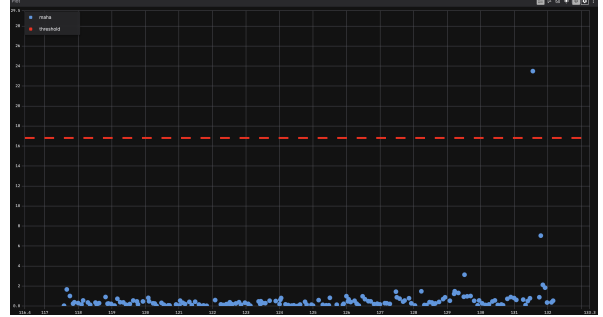


Figure 2: Mahalanobis gating: The outlier (red dashed threshold 16.81) triggers rejection, preventing a large state jump.

3 Analysis and Tuning

P/Rt Ratio. The logged ratios $\mathbf{P}_{ii}/\mathbf{R}_{t,ii}$ for position and attitude remained consistently in the 0.5–1.0 band, confirming a balanced filter where prediction and measurement are trusted similarly.

Parameters. The measurement noise was set to $\text{diag}(0.002, 0.002, 0.025, 0.004, 0.004, 0.025)$, giving horizontal ~ 1.4 cm, vertical ~ 5 cm, and attitude $\sim 2.3^\circ$ – 4.5° standard deviations. Process noise spectra in $\mathbf{Q}$ were increased to handle quick manoeuvres: gyro $2 \text{ rad/s}/\sqrt{\text{Hz}}$, accel $5 \text{ m/s}^2/\sqrt{\text{Hz}}$, gyro bias walk 5×10^{-3} , accel bias walk 5×10^{-2} . This kept innovations white and D^2 mostly within the expected χ^2 bounds while gating out occasional visual outliers.

Stability. The filter ran stably over the entire rosbag without divergence. The Euler-angle parametrisation is a potential limitation (gimbal lock at $\pm 90^\circ$ pitch), but the test trajectory never approached those angles. A quaternion-based error-state EKF would be a straightforward extension for more aggressive motions.

References

- [1] ELEC 5660 Lecture Notes, L8 – Extended Kalman Filter.